

# §11.4.128 Always-On Device Recording (Lava-side)

Classification: project-specific (the recorder target — Genymotion VM / container emulator serials — is Lava's; the §11.4.128 mandate it implements is universal, inherited via §6.AI / §6.AD).

This document records the Lava-side implementation of the HelixConstitution §11.4.128 always-on device-recording mandate, tracked under **§6.AI-debt** in the root CLAUDE.md. It states honestly what is implemented + verified versus what remains **UNVERIFIED-WITHOUT-A-DEVICE** and what is still **OWED**.

## Contract (CLAUDE.md §6.AI / §11.4.128)

A background, side-effect-free, subagent-driven recorder (logcat / perf / crash-ANR + optional screenrecord / periodic screenshot) writing raw output into the deterministic layout:

<root>/YYYY-MM-DD/<state-hash>/<DEVICE>\_<SERIAL>/recording\_NNN/

- Raw output **git-ignored + codegraph-excluded**; only **curated** evidence is ever committed.
- Default posture is **capture-and-store**; analysis happens only at release-prep or on explicit operator request — the recorder does **not** analyse.
- Capture is **read-only** w.r.t. the app under test.

## Files

| File                                       | Role                                                                                                                   |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| scripts/record-device-session.sh           | The recorder. start / stop / path subcommands. Pure path-math functions are sourceable (no side effects when sourced). |
| tests/device-recording/test_record_path.sh | Hermetic falsifiability test of the deterministic path math — <b>no device required</b> .                              |

| File                                                                                      | Role                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>.gitignore</code> → <code>.lava-ci-evidence/device-recordings/</code>               | Raw recordings git-ignored.                                                                                                                                                                                             |
| <code>.codegraph/config.json</code> → <code>.lava-ci-evidence/device-recordings/**</code> | Raw recordings excluded from the codegraph index (in addition to the pre-existing blanket <code>.lava-ci-evidence/** exclude</code> ; the explicit entry makes the intent auditable). §6.H secret excludes left intact. |

## Device serial resolution

Mirrors `scripts/run-genymotion-challenges.sh`: explicit `--serial <s>`, else the first token of `$LAVA_REAL_DEVICE_SERIALS`, else error (exit 2). The serial is expected to belong to a **Containers-submodule-driven Genymotion VM or container emulator** (§6.AG / §6.AH: never a host-direct emulator, never a borrowed live ADB device).

## Read-only guarantee

The recorder only ever runs observing adb verbs: `adb logcat -v threadtime` (**without** `-c`, which would clear the buffer — a side effect), `adb shell screenrecord`, `adb exec-out screencap`, and `adb shell getprop`. It never runs `pm clear`, `am force-stop`, or any `install/` `write` verb against the app under test.

## What is VERIFIED (no device needed)

Captured evidence — `bash tests/device-recording/test_record_path.sh`:

```
test: §11.4.128 deterministic path layout
  device-dir layout: PASS
  first index = 001: PASS
  idempotent (no creation -> same NNN): PASS
  NNN increments 001->002->003: PASS
  max+1 with gap (005 -> 006): PASS
  date override: PASS
ALL PASS — §11.4.128 path layout verified (no device required)
```

- `bash -n scripts/record-device-session.sh` → `BASH_SYNTAX_OK` (parsing clean).
- `record-device-session.sh path ... dry-run` prints the correct deterministic dir with **no device attached**, e.g. `.../2026-06-09/deadbee/Pixel-8_R5CW33CBVQV/recording_001`.

- Codegraph config remains valid JSON after the exclude insertion; all §6.H secret excludes (\*\*/.env, \*\*/keystores/\*\*, \*\*/google-services.json, \*\*/firebase-admin-\*.json, \*\*/secrets/\*\*, ...) preserved.

## Bluff-Audit (§6.N / Seventh Law clause 1)

```
Bluff-Audit: tests/device-recording/test_record_path.sh
Mutation: scripts/record-device-session.sh rds_next_index
          `printf '%03d' "$(( max + 1 ))"` → `"$(( max + 0 ))`
Observed-Failure: FAIL: first recording dir. want [.../
recording_001]
                        got [.../recording_000] (exit 1)
Reverted: yes
```

## What is UNVERIFIED-WITHOUT-A-DEVICE

The actual device-bound capture — that `adb logcat / screenrecord / screencap` against a running Genymotion VM produce non-empty, well-formed artifacts in the layout — is **NOT verified by this agent**: the Genymotion VM belongs to the operator and was not available. The `start / stop` subcommands parse clean and their `adb` invocations are read-only by construction, but **no real session has been captured**. Do **not** read this work as "it recorded a real session."

## OWED for full §11.4.128 closure (§6.AI-debt)

1. **Operator-run verification** of `start` → exercise the app on a real Containers-driven Genymotion VM → `stop`, confirming `logcat.txt / screen.mp4 / shot_*.png` are non-empty and live under the deterministic dir. Capture that run's evidence and link it here.
2. **perf / crash-ANR enrichment** beyond raw `logcat` (e.g. `dumpsys` snapshots, ANR-trace pulls from `/data/anr/`) — currently only `logcat` + screen captures.
3. **Subagent-driven always-on wiring** — auto-start the recorder around every Genymotion challenge run (hook into `scripts/run-genymotion-challenges.sh`), and a **curation step** that promotes a subset of raw recordings into committed evidence at release-prep.
4. **CM-COVENANT-114-128-PROPAGATION** literal-anchor gate (per §6.AI-debt).